

Koll. No:

BT/CS/5-26/CS6E004A/04
(Regular)



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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU & RTMNU



B. Tech Summer / Winter - 2026
(ESE / Make-up / Supplementary) Examination

BT230004CS

Branch: Computer Science & Engineering
Subject: PEC-IV (No SQL & New SQL)
Marks: 100

Sem:- VI
Subject Code: CS6E004A
Time: 3 Hrs.

Date of Exam: 02/05/2026

Instructions to the students:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand the characteristics and need for NoSQL and NewSQL systems.
2. Model, store, and query data using key-value, document, column-family, and graph databases.
3. Critically evaluate consistency, availability, and partition tolerance in NoSQL and NewSQL systems.
4. Assess performance, fault tolerance, replication, and sharding techniques in distributed databases.
5. Design and implement end-to-end mini-projects or case studies using NoSQL/NewSQL databases for applications in domains.

SECTION A

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define NoSQL database.	L1/CO1	2
Q.2	State the term Cypher query language.	L1/CO2	2
Q.3	Identify the datatypes in CQL.	L2/CO2	2
Q.4	List out examples of NewSQL databases.	L1/CO3	2
Q.5	Rewrite about HBase.	L2/CO2	2
Q.6	Classify the types of NoSQL databases.	L2/CO1	2
Q.7	Restate the term keyspace.	L2/CO2	2
Q.8	Write about fault tolerance.	L3/CO4	2
Q.9	Give examples of Key-Value databases.	L2/CO2	2
Q.10	Define aggregation.	L1/CO2	2
Q.11	State the limitations of RDBMS.	L1/CO1	2
Q.12	Enlist the types of data in NoSQL.	L1/CO1	2

SECTION B

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Distinguish between SQL and NoSQL database.	L4/CO1	4
Q.14	Illustrate with example indexing in MongoDB.	L3/CO2	4
Q.15	Discuss the need for NoSQL databases.	L2/CO1	4
Q.16	Examine Column-Family databases.	L4/CO2	4
Q.17	Write short note on Google Spanner.	L3/CO4	4
Q.18	Explain sharding and replication in details.	L3/CO4	4

SECTION C

1. Solve any FOUR Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate the CAP theorem with its implications.	L4/CO1	7
Q.20	Explain benchmarking techniques in details.	L5/CO4	7
Q.21	Analyse aggregation framework in MongoDB.	L4/CO2	7
Q.22	Compare Key-Value and Document databases.	L4/CO2	7
Q.23	Describe NewSQL systems Spanner, VoltDB, CockroachDB.	L5/CO3	7
Q.24	Discuss data modelling in Cassandra.	L5/CO2	7

SECTION D

1. Solve any TWO Questions.

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Write short note on following: a) CRUD Operation in MongoDB. b) Partitioning in Cassandra. c) Neo4j. d) Features of Cassandra.	L5/CO2	12
Q.26	Design a social network using graph database in details.	L5/CO5	12
Q.27	Interpret MongoDB architecture and operations in details.	L6/CO2	12